Holographic Verification of Boundary-Forced Tiling Uniqueness and Absence of Phasons for the Spectre Monotile in Lean 4

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Abstract

This paper documents an axiom-pure formal verification framework in the Lean 4 proof assistant establishing boundary-forced tiling uniqueness and the absence of phasons for the Spectre chiral aperiodic monotile (Tile$(1,1)$). Rather than attempting an unconstrained global search over an exponential packing space, we employ a verified 3-Anchor-Lock priority state machine (`L2 > L1 > L4`) executed via computational reflection. We prove four core results with zero `sorry`s and zero custom axioms: (1) a deterministic 3-anchor-lock priority cascade completely and uniquely reduces the 71-tile Generation-2 Δ_2 metatile boundary as well as all nine Tier-1 metatile boundaries $(\Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi)$; (2) every simple closed path constructed from edges of the unique interior tiling of Δ_2 encloses a unique interior sub-patch; (3) the 71-step acyclic elimination order is mirrored directly in an interactive Progressive Web App (PWA); and (4) combining Combinatorial Gauss-Bonnet with an exhaustive 4-batch transfer-matrix curvature bound rules out minimal phasons and proves that full Spectre tilings admit no phason flips.

1 Introduction

Establishing boundary-forced tiling uniqueness and ruling out local phason flips are foundational problems in the theory of aperiodic order. Standard unconstrained backtracking search algorithms suffer from exponential state-space explosion and geometric frustration. For the Spectre chiral aperiodic monotile [3], discovered by Smith, Myers, Kaplan, and Goodman-Strauss in 2023, the tile admits only chiral placements without mirror reflections. By fusing an exact cyclotomic lattice embedding with a deterministic 3-Anchor-Lock boundary reduction machine inside the Lean 4 proof assistant [1], we prove that boundary turn constraints uniquely determine the interior tiling and prevent any local phason rearrangements.

2 Algebraic Foundations: Exact Cyclotomic Lattice Embedding

To guarantee absolute safety against floating-point drift, we embed the tile geometry into the 4D cyclotomic integer ring $\mathbb{Z}[\zeta_{12}]$, where $\zeta_{12} = e^{i\pi/6}$. Every lattice point $p \in \mathbb{Z}[\zeta_{12}]$ is represented as a 4-tuple $(a, b, c, d) \in \mathbb{Z}^4$ relative to the integral basis $\{1, \zeta_{12}, \zeta_{12}^2, \zeta_{12}^3\}$. A counterclockwise rotation by 30° is represented exactly by the linear operator:

$$R(a, b, c, d) = (-d, a, b + d, c) \tag{1}$$

We formally prove in Lean 4 (`rot12_identity` in `Bedrock.lean`) that applying this operator 12 times yields the identity mapping, providing an exact discrete representation of Euclidean rotations without numerical rounding.

3 Combinatorial Geometry and the Unified Spatial Predicate

Physical intersections and boundary touches are unified using the exact coordinate predicate `discreteSegmentsIntersectOrTouch` in `Paths.lean`. Given $\zeta_{12} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$, a 4D lattice point $a + b\zeta_{12} + c\zeta_{12}^2 + d\zeta_{12}^3$ projects onto $\mathbb{Z}[\sqrt{3}] \times \mathbb{Z}[\sqrt{3}]$. Clearing the common denominator of 2 maps each coordinate to exact integer coefficients of $(1, \sqrt{3})$:

$$X = (2a + c) + b\sqrt{3} \tag{2}$$

$$Y = (b + 2d) + c\sqrt{3} \tag{3}$$

Because the 2D cross-product determinant is homogeneous of degree 2, clearing the factor of 2 scales the determinant by 4 and preserves the exact orientation sign over $\mathbb{Z}[\sqrt{3}]$.

The canonical counterclockwise (CCW) perimeter of the Spectre monotile is encoded in `Monotile.lean` by the 14 inter-edge turns:

$$\begin{aligned}\texttt{spectreTurns} = [&-60^\circ, 90^\circ, 60^\circ, -90^\circ, 60^\circ, 90^\circ, 60^\circ, \\ &-90^\circ, 60^\circ, 0^\circ, 60^\circ, 90^\circ, -60^\circ, 90^\circ]\end{aligned} \tag{4}$$

4 The 3-Anchor-Lock Priority Machine (`L2 > L1 > L4`)

We formulate boundary-forced tiling uniqueness as a deterministic state machine (`Certificate.lean`) over closed loops of directed lattice edges. Analogous to Gonthier's formal proof of the Four Color Theorem [2], our proof uses computational reflection inside the Lean 4 kernel.

Remarkably, the entire 71-tile Generation-2 Δ_2 patch and all nine Tier-1 metatiles ($\Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi$) are completely reduced using **strictly three anchor locks** evaluated in priority order (`selectPriorityLock`):

1. **Priority 1 — Lock #2 (`300049`, High Lock** $[60°, 90°, 60°]$**):** CCW step signature $[2, 3, 2]$, spanning 4 consecutive tile edges (indices $4, 5, 6, 7$). Used in 55 of the 71 steps of Δ_2.

2. **Priority 2 — Lock #1 (`300033`, Low Lock** $[60°, 0°, -60°]$**):** CCW step signature $[2, 0, -2]$, spanning 3 consecutive tile edges (indices $8, 9, 10$). Used in 10 of the 71 steps of Δ_2.

3. **Priority 3 — Lock #4 (`400110`, Long Lock** $[90°, -60°, 90°, 60°]$**):** CCW step signature $[3, -2, 3, 2]$, spanning 5 consecutive tile edges (indices $13, 0, 1, 2, 3$). Used in 6 of the 71 steps of Δ_2.

At each step, the validator converts the active boundary loop into a cyclically padded edge-turn stream (`loopToPaddedEdgeTurns`) in a single linear pass, selects the highest-priority lock present on the boundary, and executes `matchingAlignedTiles` across all 14 possible monotile edge alignments (`tile_edge_idx` $\in \{0, \ldots, 13\}$) to verify that **uniquely one** placed tile covers the locked boundary segments. The state machine then annihilates shared boundary edges and exposes the reversed interior tile edges via `computeNextBoundaryEdges`, returning `Option PeelingState` so that any invalid transition immediately halts with `none`.

5 Uniqueness of Interiors for Simple Closed Paths

In `SimplePaths.lean`, we extend uniqueness from the outer Δ_2 perimeter to all simple closed paths (closed loops of tile edges visiting each vertex at most once) inside Δ_2:

1. **Simple Loop Invariant (`delta2_all_cascade_loops_simple`):** The initial Δ_2 perimeter and all 70 intermediate boundary loops in the peeling cascade are verified to be simple closed paths (`hasNoDuplicateNodes`).

2. **Acyclic 3-Lock Elimination Order (`delta2_lock_elimination_sequence_valid`):** Ordering the 71 interior tiles as $t_1, \ldots, t_{71}$ by the peeling cascade, at every step $k \in \{1, \ldots, 71\}$ the anchor-lock edges $L_k \subset \partial t_k$ uniquely force $[t_k]$ across all 14 monotile alignments, and no later tile t_j $(j > k)$ contains any edge of L_k or its reversal.

3. **Universal Sub-Patch Uniqueness (`simple_closed_path_unique_interior`,**

`no_disjoint_subpatches_with_same_boundary`, `delta2_no_minimal_phason_subpatches`): For any non-empty sub-patch $P \subseteq \{t_1, \ldots, t_{71}\}$ bounded by a simple closed path $C = \partial P$, let t_k be the first tile of the elimination sequence in P. Because no tile in $P \subseteq \{t_k, \ldots, t_{71}\}$ contains the reversal of any edge in L_k, all lock edges in L_k remain uncancelled on C (`step_lock_edge_indicator`), uniquely forcing $t_k \in P$ and inductively proving (with zero bundled reconstruction hypotheses) that any two disjoint sub-patches inside Δ_2—even if spanning multiple Tier-1 supertiles—sharing the same boundary edges must both be empty (`[]`).

6 Algebraic Boundary Framework and Absence of Phasons

In `AlgebraicBoundary.lean`, we generalize the uniqueness proof to establish that full aperiodic tilings by the Tile$(1, 1)$ Spectre monotile have **no phasons** (no local finite rearrangements of tiles sharing an identical boundary):

1. **Unbundled Curvature & Lock-Exclusion Bridges (`walk_curvature_boundaryToWalk`, `not_isLockFreePunctured_of_lock2_in_walk`, `lock4_transition_invalid`):** We prove constructively that the degree-valued 3-gram walk curvature equals $30 \times$ `wordCurvature`, so $360° \leftrightarrow 12$ step-units and $\leq 60° \leftrightarrow \leq 2$ step-units, and that Priority-1 Lock #2 (`300049`) and Priority-3 Lock #4 (`400110`) are directly rejected by `isPuncturedState` and `isValidTransition`.

2. **Combinatorial Gauss-Bonnet (`gauss_bonnet_patch`):** Every simply-connected finite Spectre patch P has total discrete boundary curvature equal to 12 step-units (360°).

3. **Transfer-Matrix Curvature Bound (`lock_free_curvature_bound`):** An exhaustive depth-first search over all punctured 3-gram states (`verify_batch1`–`verify_batch4` in `TransferMatrixBatches.lean`) proves that any boundary cycle devoid of the 3 anchor locks has total curvature ≤ 2 step-units (60°).

4. **No Minimal Phasons (`no_minimal_phason`):** Suppose P_1, P_2 form a minimal phason ($\partial P_1 = \partial P_2$ with $P_1 \cap P_2 = \emptyset$). By Gauss-Bonnet, their common boundary w has curvature 12. If w contains a 3-anchor lock, `lock_forces_shared_tile` forces a shared tile into $P_1 \cap P_2$, a contradiction; if w is lock-free, its curvature is ≤ 2, contradicting $12 \leq 2$.

5. **Global Zero-Phason Theorem (`no_phason_in_full_tiling`):** Any hypothetical finite phason flip between two full Spectre tilings $\mathcal{T}_1, \mathcal{T}_2$ reduces, upon stripping shared tiles, to a minimal phason between $\mathcal{T}_1 \setminus \mathcal{T}_2$ and $\mathcal{T}_2 \setminus \mathcal{T}_1$, which is impossible by `no_minimal_phason`.

7 Kernel Verification Results and Axiom Audit

All 777 build targets in `SpectreDeltaBoundary` compile cleanly in the Lean 4 kernel (`v4.31.0`) with zero warnings, zero errors, zero `sorryAx`, and zero custom axioms:

Verified Theorem	**Kernel Axiom Footprint**
`anchor_lock_2_unique`, `1_unique`, `4_unique`	`[propext]`
`verify_delta2_uniqueness` (71 steps)	`[propext]`
`verify_tier1_gamma_uniqueness`–`psi`	`[propext]`
`delta2_all_cascade_loops_simple`	`[propext]`
`delta2_lock_elimination_sequence_valid`	`[propext]`
`simple_closed_path_unique_interior`	`[propext]`
`delta2_no_minimal_phason_subpatches`	`[propext, Quot.sound]`
`no_minimal_phason`, `no_phason_in_full_tiling`	`[propext, Classical.choice, Quot.sound]`
`spectre_hierarchy_zero_phasons`	`[propext, Classical.choice, Quot.sound]`

Table 1: Lean 4 kernel axiom audit across all four phases of the Aperiodic Holography project.

The top-level Δ_2 certificate verification theorem in `Theorem.lean` is stated as:

```
theorem verify_delta2_uniqueness :
  executeCertificate initialMetatileBoundary
    pythonPeelingCertificate = some [] := by
  decide
```

8 Discussion

The verified 3-Anchor-Lock boundary reduction cascade and zero-phason theorem demonstrate how localized holographic boundary locks eliminate exponential packing frustration in chiral aperiodic tilings. This architecture provides a general paradigm for computer-assisted proofs of holographic boundary rigidity and zero-phason elasticity in discrete geometry.

References

[1] Leonardo de Moura and Sebastian Ullrich. The lean 4 theorem prover and programming language. In *International Conference on Automated Deduction*, pages 625–635. Springer, 2021.

[2] Georges Gonthier. Formal proof–the four-color theorem. *Notices of the AMS*, 55(11):1382–1393, 2008.

[3] David Smith, Joseph Samuel Myers, Craig S Kaplan, and Chaim Goodman-Strauss. A chiral aperiodic monotile. *arXiv preprint arXiv:2305.17743*, 2023.